

ELEN 3201 Circuit Analysis Experiment 4

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1 Part 1

1.1 (a)

Given,

$$\begin{aligned} R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R_7 = R_8 &= 1\text{M}\Omega \\ C_1 = C_2 &= 1\mu\text{ F} \end{aligned}$$

we have the following

$$w_0 = \sqrt{\frac{K_1}{\tau_1 \tau_2}} = \sqrt{\frac{R_5}{R_6} \cdot \frac{1}{R_1 C_1 R_2 C_2}} = 1$$

$$\alpha = \zeta \omega_0 = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \frac{\sqrt{3}}{2}$$

$$w_0/Q = K_3/\tau_1$$

$$Q = \frac{w_0 \cdot \tau_1}{K_3} = \frac{1}{2} \Rightarrow$$

$$\xi = 0.5 < 1$$

$$2\zeta\omega_0 = 1 \Rightarrow 2\zeta(1) = 1 \Rightarrow \zeta = 0.5 \text{ so it is underdamped}$$

Initial conditions

$$V_{C_2}(0^-) = V_{C_2}(0^+) = 0$$

$$\frac{dV_{C_1}(0)}{dt} = 0$$

$$x(t)' = \frac{1}{2} e^{-t/2} \left[(\sqrt{3} B_2 - B_1) \cos\left(\frac{\sqrt{3}t}{2}\right) - (\sqrt{3} B_1 + B_2) \sin\left(\frac{\sqrt{3}t}{2}\right) \right]$$

$$(\sqrt{3}B_2 - B_1) = 0 \quad \sqrt{3} \quad B_1 = -1 \quad \Rightarrow B_2 = -\frac{1}{\sqrt{3}}$$

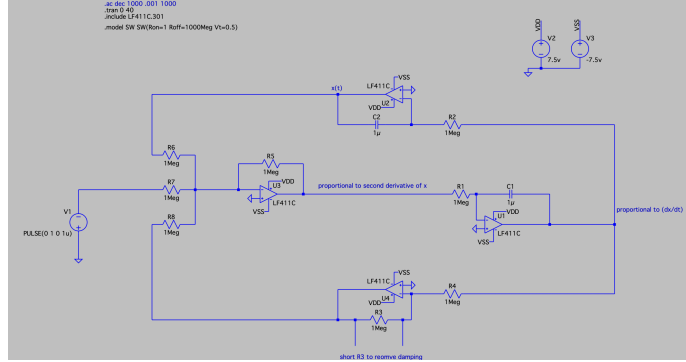


Figure 1: Part 1 Schematic

Underdamped ($\zeta < 1$ or $\alpha < \omega_0$) :

$$y_n(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

$$y = y_n + y_f = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t) + 1$$

$$s_1, s_2 = -\alpha \pm (\alpha^2 - \omega_0^2)^{1/2}$$

$$\omega_d = (\omega_0^2 - \alpha^2)^{1/2}$$

Therefore,

$$x(t) = 1 - e^{-0.5t} \left[\cos \left(\frac{\sqrt{3}}{2} t \right) + \frac{1}{\sqrt{3}} \sin \left(\frac{\sqrt{3}}{2} t \right) \right]$$

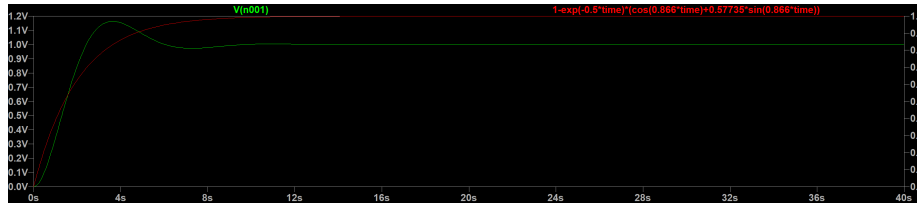


Figure 2: part 1a simulation(green) & ideal(red)

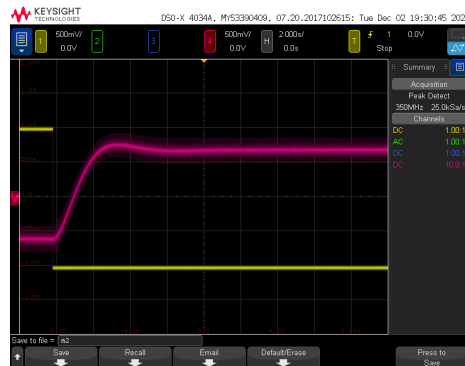


Figure 3: Part 1 a measurement

1.2 (b) shorting R_3

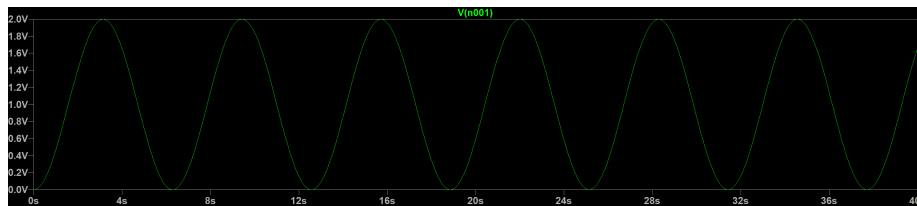


Figure 4: part1 b simulation: shorting R_3

What does shorting R_3 do to the damping factor?

The damping factor becomes zero, thus the decaying component of the solution disappears and result oscillates forever(theoratically).

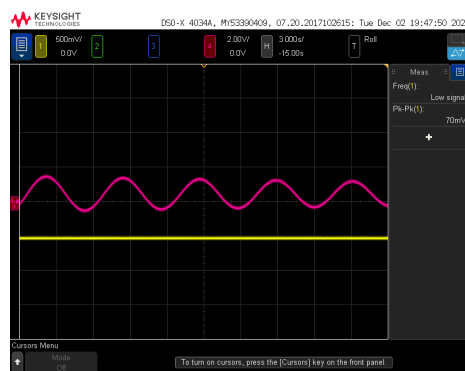


Figure 5: part b measurement

1.3 (c) R_3 making critically damped

We need $\xi = 1$

$$\zeta = 1 \Rightarrow Q = \frac{1}{2\zeta} = \frac{1}{2} = \frac{1}{K_3} \Rightarrow K_3 = 2 \Rightarrow R_3 = 2M\Omega.$$



Figure 6: part 1c Simulation Critically damped ($R_3 = 2M\Omega$)

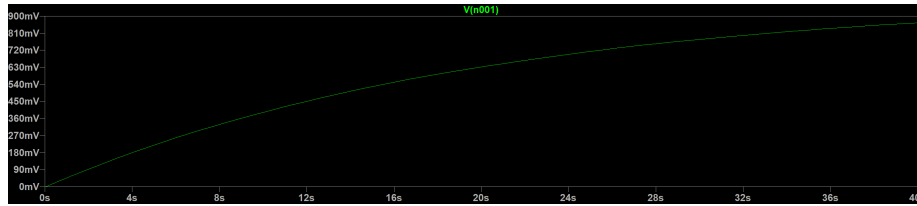


Figure 7: part1c Simulation 10x: Overdamped ($R_3 = 20M\Omega$)

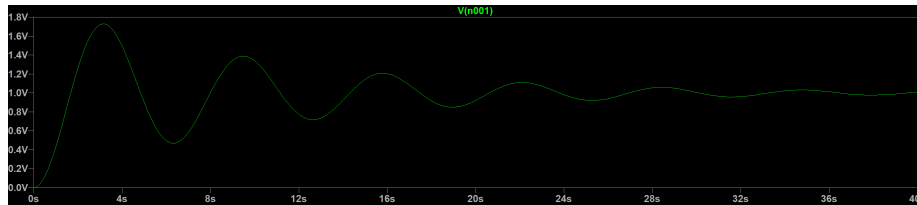


Figure 8: part 1c Simulation 0.1x: Underdamped ($R_3 = 0.2M\Omega$)



Figure 9: part 1 c overdamped measurement



Figure 10: part 1 c critically damped



Figure 11: part 1 c underdamped

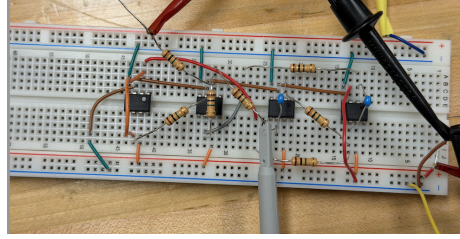


Figure 12: 1c circuit

1.4 (d) $\omega_0 = 100$

Again, $Q = \frac{1}{2}$ is needed for the system to be critically damped because $Q = \frac{1}{2\zeta} = w_0 \cdot \frac{\tau_1}{K_3} = \frac{1}{2}$. As $\omega_0 \times 100$, we need to make $K_3 = 200$ so that Q remains $\frac{1}{2}$, i.e. $100 \cdot \frac{\tau_1}{200} = \frac{1}{2}$.

Furthermore, to minimize the adjustment of the circuit, we chose the capacitor C_2 to be 10^{-4} smaller than the original to accommodate the change of K_3 while keeping C_1 the same(so τ_1 does not need to change).

$$w_0 = \sqrt{\frac{R_1}{\tau_1 \tau_2}} = \sqrt{\frac{R_5}{R_6} \cdot \frac{1}{R_1 C_1 R_2 C_2}} \Rightarrow \text{adjust } C_1, C_2$$

$$\Rightarrow C_1 \text{ not changed, } C_2 \times 10^{-4} (C_2 = 0.1 \text{ nF}) \text{ to set } \omega_0 = 100.$$

Next, picking the resistors,

$$K_3 = \frac{R_3 R_5}{R_4 R_8} = 200 \Rightarrow \frac{R_3}{R_4} = 200 \text{ (and } 2k\Omega \leq R_3 = 200R_4 \leq 10M\Omega).$$

We picked $R_4 = 2k\Omega$ and $R_3 = 400k\Omega, 40k\Omega, 4M\Omega$ (critically damped, underdamped, overdamped, respectively)

The rest of the components remain the same.

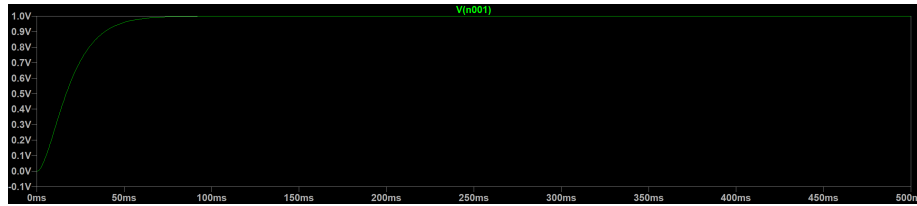


Figure 13: $R_3 = 400k$ critically damped

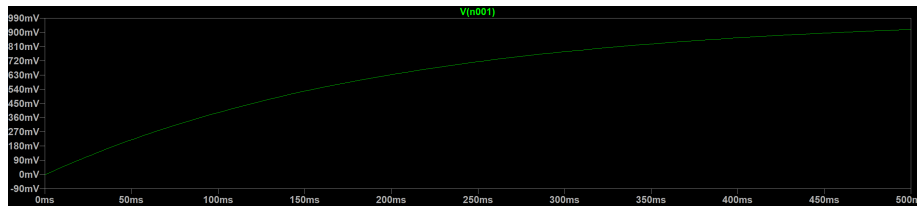


Figure 14: over damped $R3=4M\Omega$

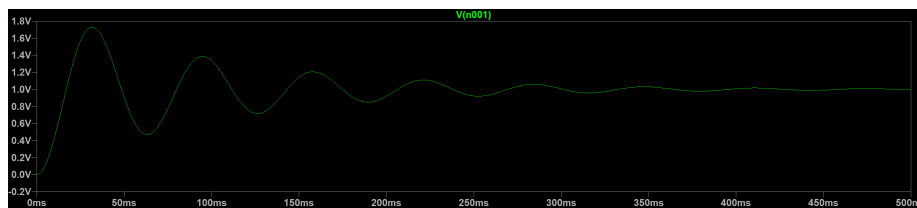


Figure 15: underdamped $R3=40k$

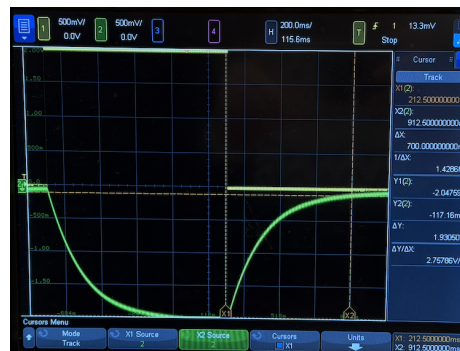


Figure 16: part 1 d overdamped

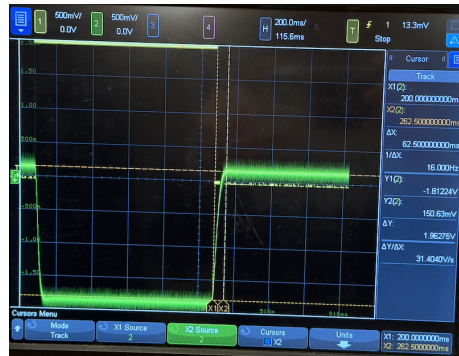


Figure 17: part1 d critically damped

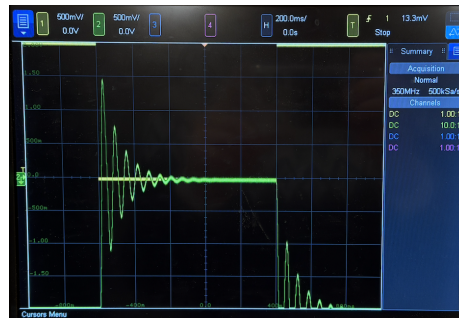


Figure 18: part 1d underdamped

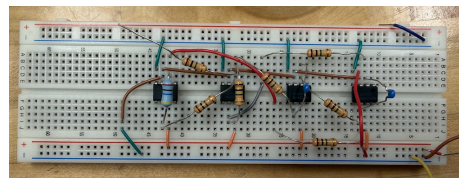


Figure 19: 1d circuit

1.5 (e) AC analysis

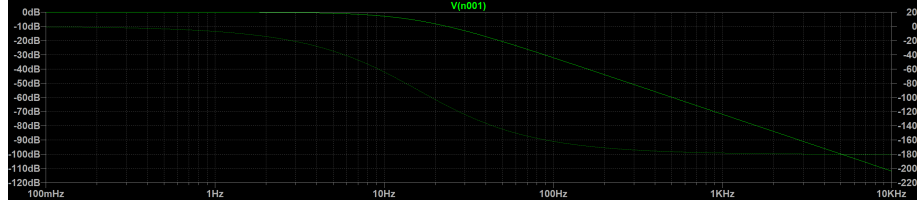


Figure 20: Part 1e AC simulation

For critically damped case ($\zeta = 1$), calculated values are :

- Low frequencies ($\omega \ll \omega_0$) : $|H(\omega)| \approx 1(0 \text{ dB})$, phase $\approx 0^\circ$
- At $\omega = \omega_0$: $|H(\omega_0)| = 0.5(-6 \text{ dB})$, phase $= -90^\circ$
- High frequencies ($\omega \gg \omega_0$) : $|H(\omega)| \approx (\omega_0/\omega)^2$, phase $\approx -180^\circ$

The simulation matches our calculated results.

1.6 (f) Q=10

Starting from part 3, we want to achieve $Q=10$. So we need to make our K_3 20 times smaller. Here, I chose $R_3 = \frac{400k}{20} \Rightarrow K_3/20$



Figure 21: Part 1f Q=10

Sharp resonance peak at $\approx 99.75\text{rad/s}$. Peak gain $\approx 10(20 \text{ dB})$ vs $0.5(-6 \text{ dB})$ for critical. Much narrower bandwidth (10 rad/s vs broad for critical). Rapid phase transition near resonance.

1.7 (g) U1 freq response output



Figure 22: U1 output

It's a bandpass filter.

Low frequencies: gain dies out $\rightarrow 0$ (stopband). At $\omega_0 = 100\text{rad/s}$: peak gain = 1(0 dB) High frequencies: gain dies out again $\rightarrow 0$ (stopband). Phase: $+90^\circ$ at low freq $\rightarrow 0^\circ$ at $\omega_0 \rightarrow -90^\circ$ at high freq. The bandwidth is about 10rad/s

1.8 (h) Verifying result

$$\tau_1 = \tau_2 = 0.01 \text{ s}$$

$$K_1 = K_2 = 1$$

$$K_3 = 0.1 \text{ (for } Q = 10 \text{ from part f)}$$

$$\omega_0 = 100\text{rad/s}$$

$$Q = 10$$

$$H_{U1}(\omega) = \frac{-j\omega \frac{1}{0.01}}{-\omega^2 + j\omega \frac{0.1}{0.01} + \frac{1}{0.01 \times 0.01}}$$

$$H_{U1}(\omega) = \frac{-j\omega \cdot 100}{-\omega^2 + j\omega \cdot 10 + 10000}$$

The U1 output represents the derivative of the state variable (proportional to dx/dt) and shows the band-pass characteristics with center frequency at ω_0 , where peak gain = 1

1.9 (i) Second response frequency

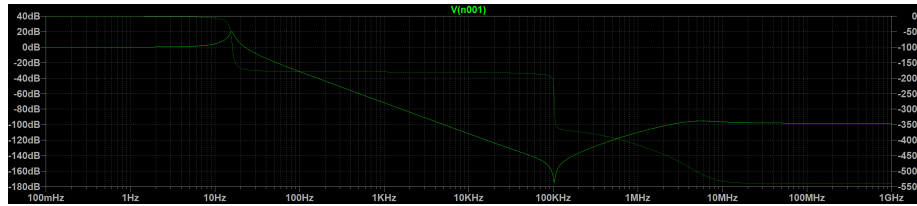


Figure 23: Extend the freq sweep range to find the second resonant freq, which is 100kH